

Appendices

Long-Term P2P Trading Among Hydrogen Refuelling Stations: An Online Optimization Method

APPENDIX A

In the dynamic cost model of liquid hydrogen storage shown in Eq. (1), the function relationship between the $f_{liquid}(S_l)$ and S_l is nonlinear. To simplify the calculation, the incremental piecewise linearization method is used to linearize the Eq. (1), expressed as follows.

$$f_{liquid}(S_l) = f_{liquid}(S_{l,1}) + \sum_{z=1}^{N_p-1} [f_{liquid}(S_{l,z+1}) - f_{liquid}(S_{l,z})] \zeta_z \quad (A1)$$

$$S_l = S_{l,1} + \sum_{z=1}^{N_p-1} [S_{l,z+1} - S_{l,z}] \zeta_z \quad (A2)$$

$$\zeta_{z+1} \leq B_z \leq \zeta_k, z = 1, 2, \dots, N_p - 2 \quad (A3)$$

$$0 \leq \zeta_z \leq 1, z = 1, 2, \dots, N_p - 1 \quad (A4)$$

where N_p represents the number of linearized segments; $S_{l,z}$ denotes the piecewise point; ζ_z reflects the position of S_l in the z -th segment interval; B_z is a binary variable; Eq. (A3) is used to ensure the continuity of the segmented interval. Similarly, Eq. (2) can be linearized using the same method, which will not be elaborated here for brevity.

APPENDIX B

The presence of the bilinear term $\pi_{buy} q_{buy}$ in Eq. (3c) introduces non-convexity into the objective function (3a). This study designs an improved segmentation-based approximation method that transforms the bilinear term $\pi_{buy} q_{buy}$ into a linear term. Improved segmentation-based approximation method divides the range of each variable into sub-regions. For each sub-region, a binary variable is defined. When the binary variable equals 1, the corresponding sub-region is activated, and a McCormick envelope is generated for linearization. The proposed method determines the minimum boundary of the non-activator region, resulting in a tighter convex envelope. The limited constraints of π_{buy} and q_{buy} are expressed as follows.

$$\pi_{buy}^{\min} \leq \pi_{buy} \leq \pi_{buy}^{\max} \quad (B1)$$

$$q_{buy}^{\min} \leq q_{buy} \leq q_{buy}^{\max} \quad (B2)$$

where $q_{buy}^{\min} = 0$. Let $m \in \{1, 2, \dots, \mathbb{M}\}$ and $n \in \{1, 2, \dots, \mathbb{N}\}$ be the partition indexes of π_{buy} and q_{buy} , respectively. For π_{buy} , let $\pi_{buy,m}^{\min}$ and $\pi_{buy,m}^{\max}$ satisfy $\pi_{buy,1}^{\min} = \pi_{buy}^{\min}$, $\pi_{buy,M}^{\max} = \pi_{buy}^{\max}$, and $\pi_{buy,m}^{\min} = \pi_{buy,m-1}^{\max}$, $\forall m \neq 1$. Likewise, for q_{buy} , let $q_{buy,n}^{\min}$ and $q_{buy,n}^{\max}$ satisfy $q_{buy,1}^{\min} = q_{buy}^{\min}$, $q_{buy,N}^{\max} = q_{buy}^{\max}$, and $q_{buy,n}^{\min} = q_{buy,n-1}^{\max}$, $\forall n \neq 1$. For any sub-region (m, n) , a binary variable $\mathcal{G}_{m,n}$ is introduced to indicate whether this sub-region is active. When $\mathcal{G}_{m,n} = 1$, the sub-region (m, n) defines the feasible region of π_{buy} and q_{buy} . In addition, an auxiliary variable ω is introduced to replace the bilinear term $\pi_{buy} q_{buy}$. Next, we will provide the mathematical details of the improved segmentation-based

approximation method. For $\pi_{buy} q_{buy}$, it has follows:

$$\sum_{m=1}^{\mathbb{M}} \sum_{n=1}^{\mathbb{N}} \mathcal{G}_{m,n} = 1 \quad (B3)$$

$$\pi_{buy} \geq \pi_{buy}^{\min} + (\pi_{buy,m}^{\min} - \pi_{buy}^{\min}) \mathcal{G}_{m,n}, \forall m, n \quad (B4)$$

$$\pi_{buy} \leq \pi_{buy}^{\max} - (\pi_{buy,m}^{\max} - \pi_{buy}^{\max}) \mathcal{G}_{m,n}, \forall m, n \quad (B5)$$

$$q_{buy} \geq q_{buy}^{\min} + (q_{buy,n}^{\min} - q_{buy}^{\min}) \mathcal{G}_{m,n}, \forall m, n \quad (B6)$$

$$q_{buy} \leq q_{buy}^{\max} - (q_{buy,n}^{\max} - q_{buy}^{\max}) \mathcal{G}_{m,n}, \forall m, n \quad (B7)$$

$$\omega \geq q_{buy,n}^{\max} \pi_{buy} + \pi_{buy,m}^{\max} q_{buy} - \pi_{buy,m}^{\max} q_{buy,n}^{\max} - a_{m,n} (1 - \mathcal{G}_{m,n}), \forall m, n \quad (B8)$$

$$\omega \geq q_{buy,n}^{\min} \pi_{buy} + \pi_{buy,m}^{\min} q_{buy} - \pi_{buy,m}^{\min} q_{buy,n}^{\min} - b_{m,n} (1 - \mathcal{G}_{m,n}), \forall m, n \quad (B9)$$

$$\omega \leq q_{buy,n}^{\min} \pi_{buy} + \pi_{buy,m}^{\max} q_{buy} - \pi_{buy,m}^{\max} q_{buy,n}^{\min} - c_{m,n} (1 - \mathcal{G}_{m,n}), \forall m, n \quad (B10)$$

$$\omega \leq q_{buy,n}^{\max} \pi_{buy} + \pi_{buy,m}^{\min} q_{buy} - \pi_{buy,m}^{\min} q_{buy,n}^{\max} - d_{m,n} (1 - \mathcal{G}_{m,n}), \forall m, n \quad (B11)$$

$$a_{m,n} = \pi_{buy}^{\max} q_{buy}^{\max} - \pi_{buy,m}^{\max} q_{buy,n}^{\max} - (q_{buy}^{\max} - q_{buy,n}^{\max}) \pi_{buy}^{\min} - (\pi_{buy}^{\max} - \pi_{buy,m}^{\max}) q_{buy}^{\min}, \forall m, n \quad (B12)$$

$$b_{m,n} = \pi_{buy}^{\min} q_{buy}^{\min} - \pi_{buy,m}^{\min} q_{buy,n}^{\min} - (q_{buy}^{\min} - q_{buy,n}^{\min}) \pi_{buy}^{\max} - (\pi_{buy}^{\min} - \pi_{buy,m}^{\min}) q_{buy}^{\max}, \forall m, n \quad (B13)$$

$$c_{m,n} = \pi_{buy}^{\max} q_{buy,n}^{\min} - \pi_{buy,m}^{\max} q_{buy}^{\min} + (q_{buy}^{\min} - q_{buy,n}^{\min}) \pi_{buy}^{\min} - (\pi_{buy}^{\max} - \pi_{buy,m}^{\max}) q_{buy}^{\max}, \forall m, n \quad (B14)$$

$$d_{m,n} = \pi_{buy,m}^{\min} q_{buy,n}^{\max} - \pi_{buy}^{\min} q_{buy}^{\max} + (q_{buy}^{\max} - q_{buy,n}^{\max}) \pi_{buy}^{\max} - (\pi_{buy}^{\min} - \pi_{buy,m}^{\min}) q_{buy}^{\min}, \forall m, n \quad (B15)$$

Eqs. (B1)-(B15) form a convex envelope within each sub-region, and the bilinear term $\pi_{buy} q_{buy}$ is represented by a linear auxiliary variable ω . This completes the linear approximation of the bilinear term $\pi_{buy} q_{buy}$. Similarly, the bilinear term $\pi_{sell} q_{sell}$ can be linearized using the same approach, which will not be reiterated here.

APPENDIX C

Applying strong duality theory, the supremum problem in Eq. (16a) can be transformed into an infimum problem, expressed as follows.

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}_{\mathbb{P}} [f_{\text{spot}}(\mathbf{x}, \xi)] = \inf_{\lambda \geq 0} \left\{ \frac{1}{N} \sum_{o=1}^N \sup_{\xi \in \Xi} \left(f_{\text{spot}}(\mathbf{x}, \xi) - \lambda \|\hat{\xi}_o - \xi\|_1 \right) + \lambda \ell \right\} \quad (\text{C1})$$

where λ denotes the dual variable. The objective function and constraints of the DRO operation in the forward market can be rewritten as:

$$\min_{\mathbf{x}} \mathbf{a}^T \mathbf{x} + \lambda \ell + \frac{1}{N} \sum_{o=1}^N \sup_{\xi \in \Xi} \left(f_{\text{spot}}(\mathbf{x}, \xi) - \lambda \|\hat{\xi}_o - \xi\|_1 \right) \quad (\text{C2})$$

$$\text{s.t. } \mathbf{Ax} \leq \mathbf{c} \quad (\text{C3})$$

$$\mathbf{Gy}(\mathbf{x}, \xi) \leq \mathbf{h}(\xi) \quad (\text{C4})$$

By introducing an auxiliary variable $\Psi_o (\forall o=1, 2, \dots, N)$, the above min-sup problem can be further transformed into:

$$\min_{\mathbf{x}} \mathbf{a}^T \mathbf{x} + \lambda \ell + \frac{1}{N} \sum_{o=1}^N \Psi_o \quad (\text{C5})$$

$$\text{s.t. } \sup_{\xi \in \Xi} \left(f_{\text{spot}}(\mathbf{x}, \xi) - \lambda \|\hat{\xi}_o - \xi\|_1 \right) \leq \Psi_o \quad (\text{C6})$$

$$\mathbf{Ax} \leq \mathbf{c} \quad (\text{C7})$$

$$\mathbf{Gy}(\mathbf{x}, \xi) \leq \mathbf{h}(\xi) \quad (\text{C8})$$

Based on the DRO operation model of HRS in the forward market, $f_{\text{spot}}(\mathbf{x}, \xi)$ is a convex function concerning ξ . Therefore, optimal solution of $\sup_{\xi \in \Xi} (f_{\text{spot}}(\mathbf{x}, \xi) - \lambda \|\hat{\xi}_o - \xi\|_1)$ must be attained at the upper bound $\hat{\xi}_o$, the lower bound $\underline{\xi}$, or the point $\hat{\xi}_o \in \Xi$ of ξ . Furthermore, $\mathbf{y}(\mathbf{x}, \xi)$ and $\mathbf{h}(\xi)$ are linear functions concerning ξ . Within the support set Ξ , the optimal solution of $\sup_{\xi \in \Xi} (f_{\text{spot}}(\mathbf{x}, \xi) - \lambda \|\hat{\xi}_o - \xi\|_1)$ must be found at the upper or lower bounds of ξ . Thus, Eqs. (C5)-(C8) are reformulated as a directly solvable MILP problem, shown in Eq. (17).

The proof is completed. ■

APPENDIX D

PROOF OF THEOREM 1

The core of the proof is to demonstrate that the control system's value function $V(e_t)$ is an ISS-Lyapunov function for the tracking error e_t .

Step 1: Constructing the candidate Lyapunov function

Define the value function as the candidate Lyapunov function, as shown in Eq. (D1).

$$V(e_t) = \min_{u \in \mathcal{U}} J(x_t^H, r_t, u) = \min_{u \in \mathcal{U}} \sum_{\tau=t}^{t+\pi} \left[C(x_\tau^H, u_\tau) + \alpha \|x_\tau^H - r_\tau\|^2 \right] \quad (\text{D1})$$

where the sequences $\{x_\tau^H\}$ and $\{u_\tau\}$ satisfy the system dynamics $x_{\tau+1}^H = f_h(x_\tau^H, u_\tau)$.

Step 2: Proving positive definiteness and radial unboundedness of the Lyapunov function

Since $C(x^H, u) \geq 0$ and $\alpha > 0$, we have $V(e_t) \geq 0$ and $V(0) = 0$. Due to the strong convexity of the cost function and the compact constraint set, there exist class $\mathcal{K}_\infty$ functions $\underline{\alpha}$ and $\bar{\alpha}$ such that:

$$\underline{\alpha}(\|e_t\|) \leq V(e_t) \leq \bar{\alpha}(\|e_t\|) \quad (\text{D2})$$

Because $C(x^H, u)$ is strongly convex, minimum of $V(e_t)$ grows at least quadratically on the compact set, implying the existence of $\underline{m}, \bar{M} > 0$ such that $\underline{m}\|e_t\|^2 \leq V(e_t) \leq \bar{M}\|e_t\|^2$. Set

$\underline{\alpha}(s) = \underline{m}s^2$ and $\bar{\alpha}(s) = \bar{M}s^2$. This ensures that $V(e_t)$ is positive definite and radially unbounded.

Step 3: Proving the decay property of Lyapunov function

We need to prove that for any e_t , there exists a class $\mathcal{K}_\infty$ function α_3 and a class $\mathcal{K}$ function σ such that:

$$V(e_{t+1}) - V(e_t) \leq -\alpha_3(\|e_t\|) + \sigma(\|r_{t+1} - r_t\| + \|w_t\|) \quad (\text{D3})$$

The derivation is as follows:

a) Constructing a feasible control sequence

Let $u_t^* = \{u_t^*, u_{t+1}^*, \dots, u_{t+\pi}^*\}$ be the optimal solution to the optimization problem at time t , corresponding to the state sequence $x_t^{*H} = \{x_t^{*H}, x_{t+1}^{*H}, \dots, x_{t+\pi}^{*H}\}$. At time $t+1$, we construct a feasible but not necessarily optimal control sequence $\tilde{u} = \{u_{t+1}^*, u_{t+2}^*, \dots, u_{t+\pi}^*, u_f\}$, where u_f is a feasible terminal control law (e.g., zero input). Let $\tilde{x} = \{\tilde{x}_{t+1}^H, \tilde{x}_{t+2}^H, \dots, \tilde{x}_{t+\pi+1}^H\}$ be the corresponding state sequence, $\tilde{x}_{t+1}^H = x_{t+1}^{*H} = f_h(x_t^H, u_t^*) + w_t$.

b) Calculating the difference in value functions

Since $\tilde{u}$ is feasible at time $t+1$ but not necessarily optimal, we have that

$$V(e_{t+1}) \leq J(x_{t+1}^H, r_{t+1}, \tilde{u}) \quad (\text{D4})$$

The difference in value functions is expressed as follows:

$$\begin{aligned} V(e_{t+1}) - V(e_t) &\leq J(x_{t+1}^H, r_{t+1}, \tilde{u}) - J(x_t^H, r_t, u_t^*) \\ &\leq \left[\sum_{\tau=t+1}^{t+\pi} \left(C(\tilde{x}_\tau^H, \tilde{u}_\tau) + \alpha \|\tilde{x}_\tau^H - r_\tau\|^2 \right) \right] - \\ &\quad \left[\sum_{\tau=t}^{t+\pi} \left(C(x_\tau^{*H}, u_\tau^*) + \alpha \|x_\tau^{*H} - r_\tau\|^2 \right) \right] \end{aligned} \quad (\text{D5})$$

c) Decomposition and simplification

For $\tau = t+1, \dots, t+\pi$, we have $\tilde{u}_\tau = u_\tau^*$. However, due to disturbance w_t and reference variation $r_{t+1} - r_t$, the corresponding state sequences $\tilde{x}_\tau^H$ and x_τ^{*H} differ. Rewriting the above equation yields:

$$V(e_{t+1}) - V(e_t) \leq - \left[C(x_t^{*H}, u_t^*) + \alpha \|e_t\|^2 \right] +$$

$$\sum_{\tau=t+1}^{t+\pi} \left[C(\tilde{x}_\tau^H, u_\tau^*) - C(x_\tau^{*H}, u_\tau^*) + \alpha (\|\tilde{x}_\tau^H - r_\tau\|^2 - \|x_\tau^{*H} - r_\tau\|^2) \right] \quad (\text{D6})$$

Given the Lipschitz continuity of f_h and C , and the boundedness of the reference variation, there exist constants $L_f, L_c > 0$ such that:

$$\|\tilde{x}_\tau^H - x_\tau^{*H}\| \leq L_f^{\tau-t-1} (\|w_t\| + L_\Phi \Delta_r), \quad \text{for } \tau \geq t+1 \quad (\text{D7})$$

$$|C(\tilde{x}_\tau^H, u_\tau^*) - C(x_\tau^{*H}, u_\tau^*)| \leq L_c \|\tilde{x}_\tau^H - x_\tau^{*H}\| \quad (\text{D8})$$

$$\|\tilde{x}_\tau^H - r_\tau\|^2 - \|x_\tau^{*H} - r_\tau\|^2 \leq 2\bar{R} \|\tilde{x}_\tau^H - x_\tau^{*H}\| \quad (\text{D9})$$

where $\bar{R}$ denotes the upper bound on the magnitude of state and reference deviation.

Therefore, the sum of the difference terms can be bounded as follows:

$$\begin{aligned} &\sum_{\tau=t+1}^{t+\pi} \left[C(\tilde{x}_\tau^H, u_\tau^*) - C(x_\tau^{*H}, u_\tau^*) + \alpha (\|\tilde{x}_\tau^H - r_\tau\|^2 - \|x_\tau^{*H} - r_\tau\|^2) \right] \\ &\leq L' (\|w_t\| + \Delta_r) \end{aligned} \quad (\text{D10})$$

where $L' = (L_c + 2\alpha\bar{R}) \sum_{k=0}^{\pi-1} L_f^k$.

d) Final decay inequality

Since $C(x^H, u)$ is μ -strongly convex and the penalty term $\alpha \|e_t\|^2$ is quadratic, there exists μ' such that:

$$C(x_t^{*H}, u_t^*) + \alpha \|e_t\|^2 \geq \mu' \|e_t\|^2 \quad (D11)$$

where μ' depends on μ , α , and the sensitivity of the system dynamics. Under the sufficient stability condition $\alpha > \alpha_{\min}$, we have $\mu' > 0$.

Combining the above analysis, we can obtain that

$$V(e_{t+1}) - V(e_t) \leq -\mu' \|e_t\|^2 + L'(\|w_t\| + \Delta_r) \quad (D12)$$

Let $\alpha_3 = \mu's^2$ (a class $\mathcal{K}_\infty$ function) and $\sigma(s) = L's$ (a class $\mathcal{K}$ function). Then,

$$V(e_{t+1}) - V(e_t) \leq -\alpha_3(\|e_t\|) + \sigma(\|w_t\| + \|r_{t+1} - r_t\|) \quad (D13)$$

This proves the decay property of the Lyapunov function.

Step 4: Deriving ultimate boundedness

From Eq. (D12), when $\|e_t\|$ is sufficiently large, $V(e_t)$ decreases. Therefore, the system state will eventually enter the set $\{\|e_t\| \leq B\}$, where B satisfies $\mu'B^2 = L'(\|w_t\| + \Delta_r)$. Since this bound may be loose, we derive a tighter bound from the error dynamics. Considering the controller performance, there exists a constant $\Upsilon \in (0, 1)$ such that the closed-loop system satisfies the following:

$$\|e_{t+1}\| \leq \Upsilon \|e_t\| + \|r_{t+1} - r_t\| + \|w_t\| \quad (D14)$$

This inequality can be derived from the first-order optimality condition of the optimization problem. At steady state, it has $\|e_{t+1}\| = \|e_t\| = \bar{e}$. Substituting into Eq. (D12) yields:

$$\bar{e} \leq \Upsilon \bar{e} + \Delta_r + W \quad (D15)$$

Thus, we obtain that

$$\bar{e} \leq \frac{\Delta_r + W}{1 - \Upsilon} \quad (D16)$$

This is the ultimate bound for the tracking error. To ensure $\bar{e} \leq B_{\max}$, we need:

$$\Delta_r \leq (1 - \Upsilon) B_{\max} - W \quad (D17)$$

This naturally leads to the upper bound $\Delta_{r, \max}$ on the reference variation rate in the proof process.

Step 5: Derivation of sufficient conditions for stability

(a) Derivation of the lower bound $\alpha_{\min}$ of the penalty coefficient

From the proof of the Lyapunov function's decay, we require $\mu' > 0$. By analyzing the Hessian matrix of the optimization problem, the minimum eigenvalue satisfies as follows:

$$\lambda_{\min}(\nabla^2 J) \geq \mu + 2\alpha\lambda_{\min}(M) \quad (D18)$$

where M is the sensitivity matrix of the system dynamics. To ensure strong convexity, we need $\mu + 2\alpha\lambda_{\min}(M) \geq L_f^2$, and it has following:

$$\alpha > \frac{L_f^2 - \mu}{2\lambda_{\min}(M)} \quad (D19)$$

Considering the Lipschitz constant L_c of the cost function, a more conservative bound $\alpha_{\min} = (L_c + L_f)/\mu$ is adopted to ensure the strong convexity of the objective function J is sufficient to resist disturbances.

(b) Derivation of the upper bound $\Delta_{r, \max}$ for the reference variation rate.

As stated in Eq. (D17), it has $\Delta_r \leq (1 - \Upsilon) B_{\max} - W$.

In summary, the proof of Theorem 1 is completed. ■

APPENDIX E

PROOF OF THEOREM 2

(a) Convergence of the reference generation subsystem

Under the following conditions:

- 1) The uncertainty process $\{w_t\}$ is stationary and ergodic.
- 2) The bandwidth parameter $\sigma > 0$ is chosen appropriately.
- 3) The number of historical scenarios S is sufficiently large.

Then, the reference trajectory r_t converges almost surely:

$$\lim_{t \rightarrow \infty} r_t = r_\infty = \mathbb{E}[r^* | \xi_{[t]}] \quad (E1)$$

Proof:

Step 1: Consistency of kernel regression

By Nadaraya-Watson estimation theory, for appropriate kernel functions and bandwidth, we obtain follows.

$$\sup_{x^H \in \mathcal{X}} \left| \Phi(\xi_{[t]}) - \mathbb{E}[r^* | \xi_{[t]} = x^H] \right| \rightarrow 0 \quad (E2)$$

Step 2: Convergence of weights

Define the true posterior probability expressed as follows.

$$p_s^* = \mathbb{P}(s | \xi_{[t]}) \quad (E3)$$

By the law of large numbers, it has $\rho_{s,t} \rightarrow p_s^*$.

Step 3: Convergence of the reference trajectory

Therefore, we can derive the following:

$$r_t = \sum_{s=1}^S \rho_{s,t} r_s^* \rightarrow \sum_{s=1}^S p_s^* r_s^* = \mathbb{E}[r^* | \xi_{[t]}] \quad (E4)$$

This completes the proof of convergence of the reference generation subsystem.

(b) Kernel bandwidth selection

Kernel bandwidth parameter σ is determined as follows.

$$\sigma = O(S^{-1/5}) \quad (E5)$$

Proof:

The selection of the kernel bandwidth σ is based on kernel regression theory to minimize the Mean Square Error (MSE) of the reference estimate.

Step 1: Kernel regression model

The reference generation is $r_t = \Phi(\xi_{[t]}) = \sum_{s=1}^S \rho_{s,t} r_s^*$, where the weights $\rho_{s,t}$ are calculated by the Gaussian kernel function, expressed as follows.

$$\rho_{s,t} \propto \left(-\frac{\|\xi_{[t]} - \xi_{[t],s}\|_2^2}{2\sigma^2 S} \right) \quad (E6)$$

Step 2: MSE decomposition

The MSE consists of bias and variance, as follows.

$$MSE = Bias^2 + Variance \quad (E7)$$

The *Bias* increases with σ ; the *Variance* decreases with σ .

Step 3: Optimal bandwidth selection

For Gaussian kernels and one-dimensional data, the optimal bandwidth σ relates to the sample size S as:

$$\sigma \propto S^{-1/5} \quad (E8)$$

This result comes from the asymptotic analysis of minimizing MSE. Specifically, by solving:

$$\frac{\partial \text{MSE}}{\partial \sigma} = 0 \quad (E9)$$

We obtain that $\sigma = \tilde{c} \cdot S^{-1/5}$, where $\tilde{c}$ is a constant depending on the data distribution.

To ensure convergence of the reference trajectory, we select $\sigma = O(S^{-1/5})$, which balances bias and variance, making the reference estimate converge consistently to the true conditional expectation $\mathbb{E}[r^* | \xi_{[t]}]$.

(c) Convergence of the composite system

Proof:

Step 1: Application of reference convergence

From Eq. (E1), we have $r_t \rightarrow r_\infty$.

Step 2: Convergence of tracking error

Define the limiting tracking control system as follows.

$$\begin{cases} x_{t+1}^H = f_h(x_t^H, \kappa(x_t^H, r_\infty)) + w_t \\ r_{t+1} = r_\infty \end{cases} \quad (E10)$$

This is a standard reference tracking problem. By predictive control stability theory, there exists an invariant set $\mathcal{X}_\infty$ such that:

$$\mathcal{X}_\infty = \{(x^H, r) : \|x^H - r\| \leq B, r \in \mathcal{R}_\infty\} \quad (E11)$$

where $\mathcal{R}_\infty$ denotes the limit set of the reference trajectory.

Step 3: Composite convergence:

Since $r_t \rightarrow r_\infty$ and e_t is bounded, therefore, it has

$$\lim_{t \rightarrow \infty} D_{\text{dist}}((x_t^H, r_t), \mathcal{A}) = 0 \quad (E12)$$

This completes the proof of Theorem 2. $\blacksquare$

APPENDIX F

(1) Preference attribute design

This study constructs the preference attribute list of HRS from four indicators: price preference, distance preference, reputation preference, and future value preference.

(a) *Price preference*:

Buyer HRSs prefer lower purchase prices, while sellers prefer higher selling prices. Therefore, when calculating a buyer's preference for a seller, a lower seller's asking price increases buyer preference; conversely, a higher buyer's bid price increases seller preference for that buyer.

(b) *Distance preference*:

A shorter geographical distance dist_{ij} leads to lower transportation costs. Both buyer and seller HRS prefer trading parties that are geographically closer.

(c) *Reputation preference*:

HRSs with good reputations from historical trading are more favored. This study uses the historical order fulfillment rate to describe reputation preference, which is expressed as follows.

$$\Theta_i^{\text{rep}} = \sum_{t \in T} \Gamma_{i,t} / \sum_{t \in T} \Lambda_{i,t} \quad (F1)$$

where Θ_i^{rep} represents the historical order fulfillment rate; $\Gamma_{i,t}$ denotes the actual successfully delivered volume (the sum of all completed purchases or sales of HRS i during the trading

period t); $\Lambda_{i,t}$ represents the total contracted volume (the sum of all agreed-upon transactions of HRS i during the trading period t , regardless of final fulfillment). A ratio closer to 1 indicates higher reputation.

(d) *Future value preference*:

The future value preference is defined as follows.

$$\Delta V_{\text{value},ij} = \frac{\tilde{\gamma} \Delta V_{\text{value},i}}{V_{\text{value},i}^{\text{base}}} + \frac{\tilde{\gamma} \Delta V_{\text{value},j}}{V_{\text{value},j}^{\text{base}}} \quad (F2)$$

where $\Delta V_{\text{value},i}$ denotes the value adjustment for buyer i , with $\Delta V_{\text{value},i} = V_{\text{value},i}(\mathcal{S}_i^{\text{post}}) - V_{\text{value},i}(\mathcal{S}_i^{\text{pre}})$; $V_{\text{value},i}(\mathcal{S}_i^{\text{post}})$ is the post-transaction value; $V_{\text{value},i}(\mathcal{S}_i^{\text{pre}})$ is the current state value; $\Delta V_{\text{value},j}$ denotes the value adjustment for seller j , with $\Delta V_{\text{value},j} = V_{\text{value},j}(\mathcal{S}_j^{\text{post}}) - V_{\text{value},j}(\mathcal{S}_j^{\text{pre}})$; $V_{\text{value},j}(\mathcal{S}_j^{\text{post}})$ is the post-transaction value; $V_{\text{value},j}(\mathcal{S}_j^{\text{pre}})$ is the current state value; $V_{\text{value},i}^{\text{base}}$ is the baseline value, represented by the average daily benefit of the HRS.

As the calculated preference indicators may have different scales, each indicator is normalized to the range [0,1] by dividing it by the maximum value of that indicator to eliminate dimensionality. The preference score is defined as the weighted sum of the above preference attributes.

Preference list definition: Each buyer or seller can construct a preference list for the other side, ranked in strict descending order according to the preference score calculated by Eq. (34).

For example, for seller sl_j , its preference ranking relation $\succ_{sl_j}$ is defined as:

$$by_i \succ_{sl_j} by_{i'} \Leftrightarrow \text{Score}_{sl_j \rightarrow by_i} > \text{Score}_{sl_j \rightarrow by_{i'}} \quad (F3)$$

Eq. (F3) indicates that seller sl_j prefers buyer $by_{i'}$ over buyer by_i . Thus, each seller can use this preference relation to rank all buyers in strict descending order, constructing a seller preference list $\mathcal{P}_{\mathcal{L}_{sl_j}}$.

Similarly, for buyer by_i , its preference ranking relation $\succ_{by_i}$ is defined as:

$$sl_j \succ_{by_i} sl_{j'} \Leftrightarrow \text{Score}_{by_i \rightarrow sl_j} > \text{Score}_{by_i \rightarrow sl_{j'}} \quad (F4)$$

Each buyer HRS can form its own preference list $\mathcal{P}_{\mathcal{L}_{by_i}}$ by ordering all sellers in decreasing order.

(2) Matching algorithm procedure

First, construct preference lists $\mathcal{P}_{\mathcal{L}_{by_i}}$ and $\mathcal{P}_{\mathcal{L}_{sl_j}}$ for each buyer by_i and seller sl_j . Initialize all buyers and sellers as unmatched (partial matching is allowed, i.e., the matched volume can be less than total demand/supply). Each buyer HRS, following its preference list order, sequentially sends trade requests to sellers, with the request amount being its remaining demand. Upon receiving a request, if the seller's remaining supply is greater than 0, it considers whether to accept the request. The seller selects its most preferred buyer according to its own preference list order until its supply is allocated. If a buyer's request is rejected, the buyer selects its most preferred seller among those who have not yet rejected it and proposes its current trade request. These processes repeat until all buyer demands are satisfied or all sellers' hydrogen supply is allocated.

Theorem 3: The matching result obtained by the proposed many-to-many matching algorithm is a stable matching.

Proof: Using proof by contradiction, assume the proposed algorithm does not yield a stable matching. Then there exists a blocking pair (sl_i, by_i^*) satisfying $by_i^* \succ_{sl_i} \mathcal{M}(sl_i)$ and $sl_i \succ_{by_i^*} \mathcal{M}(by_i^*)$. Based on $sl_i \succ_{by_i^*} \mathcal{M}(by_i^*)$, before applying to $\mathcal{M}(by_i^*)$, buyer by_i^* would have prioritized sending a request to sl_i but was rejected, then proceeded to its next preferred seller HRS. Therefore, $\mathcal{M}(sl_i) \succ_{sl_i} by_i^*$. This contradicts the definition of a blocking pair. Hence, the assumption is false, proving that the achieved matching result is stable. ■

APPENDIX G

The geographical distance distances among HRSs are shown in Table G1.

TABLE G1 GEOGRAPHICAL DISTANCES AMONG HRSs

	HRS1	HRS2	HRS3	HRS4	HRS5	HRS6	HRS7	HRS8	HRS9	HRS10
HRS1	0	19km	39km	8.6km	75km	59km	71km	50km	18km	20km
HRS2	19km	0	47km	24km	60km	44km	57km	26km	2km	21km
HRS3	39km	47km	0	42km	88km	72km	85km	39km	45km	28km
HRS4	8.6km	24km	42km	0	79km	64km	76km	54km	22km	30km
HRS5	75km	60km	88km	79km	0	15km	3km	54km	54km	60km
HRS6	59km	44km	72km	64km	15km	0	12km	42km	42km	45km
HRS7	71km	57km	85km	76km	3km	12km	0	63km	59km	58km
HRS8	50km	26km	39km	54km	54km	42km	63km	0	35km	18km
HRS9	18km	2km	45km	22km	54km	42km	59km	35km	0	20km
HRS10	20km	21km	28km	30km	60km	45km	58km	18km	20km	0

The installed equipment capacities within HRSs are shown in Table G2.

TABLE G2 INSTALLED EQUIPMENT CAPACITIES OF HRSs

HRS	GHS	LHS	SHS	WT	PV	ES	EL	CP
HRS1	1000kg	1000kg	0kg	2000kW	2000kWp	2000kWh	2000kW	500kW
HRS2	1000kg	1000kg	0kg	3000kW	1500kWp	1500kWh	2500kW	500kW
HRS3	1500kg	1000kg	0kg	2500kW	1500kWp	1500kWh	2000kW	500kW
HRS4	1500kg	1000kg	0kg	1500kW	2500kWp	1000kWh	2500kW	500kW
HRS5	1000kg	0kg	1000kg	1500kW	1500kWp	2000kWh	2000kW	500kW
HRS6	1000kg	0kg	1000kg	1500kW	1800kWp	2200kWh	2000kW	500kW
HRS7	500kg	0kg	1000kg	2000kW	1500kWp	500kWh	2000kW	500kW
HRS8	1000kg	0kg	1000kg	1500kW	1500kWp	1000kWh	2000kW	500kW
HRS9	1500kg	1000kg	0kg	2000kW	2000kWp	2000kWh	2000kW	500kW
HRS10	1000kg	0kg	1000kg	1500kW	1500kWp	1000kWh	2000kW	500kW

The economic and technical parameters of the equipment are shown in Table G3.

TABLE G3 PARAMETERS OF HRS's EQUIPMENT

Parameter	Value	Parameter	Value	Parameter	Value
$\beta_{liquid,om}$	0.005 ¥ /kg	η_{liquid}	1.2	λ_{liquid}	0.0001/h
ζ_{liquid}	0.05 ¥ /kg	γ_{solid}	0.08 ¥ /kg	$\delta_{solid,om}$	0.01 ¥ /kg
θ_{solid}	0.8	μ_{solid}	0.0001/h	LHV_{H_2}	33.3kWh/kg
η_{ele}^{rated}	0.02	$\tilde{\kappa}$	1.4	R_{H_2}	14.304kJ/(kg*K)
α_{ele}	0.001	H_{in}	10Mpa	T_{in}	293 K
ΔP_{ele}^{max}	1000kW	H_{out}	45Mpa	Q_{comp}^{max}	500kW
P_{ele}^{min}	0kW	P_{ele}^{max}	Cap _{EL}	η_{com}	0.54
ζ_{LHS}	0.001	$c_{com,wind}$	0.1 ¥ /kW	$c_{com,pv}$	0.08 ¥ /kW
$c_{com,ele}$	0.2 ¥ /kW	$c_{stor,LHS}$	0.005 ¥ /kg	$c_{stor,SHS}$	0.01 ¥ /kg
$c_{com,com}$	0.15 ¥ /kW	$c_{stor,es}$	0.001 ¥ /kW	$c_{stor,GHS}$	0.001 ¥ /kg
$u_{GHS,in}^{max}, u_{GHS,out}^{max}$	50kg, 50kg	$u_{es,char}^{max}$	0.2* Cap _{ES}	$u_{es,dischar}^{max}$	0.2* Cap _{ES}

$\eta_{es,dischar}$	$\eta_{es,char}$	0.95, 0.95	$u_{SHS,in}^{max}, u_{SHS,out}^{max}$	20kg, 20kg	$u_{LHS,in}^{max}, u_{LHS,out}^{max}$	20kg, 20kg
S_{es}^{min}		0.1*Cap _{ES}	S_{GHS}^{max}	0.95*Cap _{GHS}	S_{LHS}^{max}	0.9*Cap _{LHS}
S_{es}^{max}		0.9*Cap _{ES}	S_{SHS}^{min}	0.01*Cap _{SHS}	S_{SHS}^{max}	0.9*Cap _{SHS}
$P_{wind}^{max}, P_{pv}^{max}$		Cap _{WT} , Cap _{PV}	S_{GHS}^{min}	0.05*Cap _{GHS}	S_{LHS}^{min}	0.01*Cap _{LHS}
$\eta_{LHS,in}, \eta_{LHS,out}$		0.98, 0.98	$\eta_{GHS,in}, \eta_{GHS,out}$	0.98, 0.98	$\eta_{SHS,in}, \eta_{SHS,out}$	0.98, 0.98

In Table G3, “Cap” represents the installed equipment capacity corresponding to the subscript. For example, “Cap_{GHS}” represents the installed capacity of GHS, as shown in Table G2.

The parameters of DQN are shown in Table G4.

TABLE G4 NEURAL NETWORK STRUCTURE AND PARAMETERS OF DQN

Parameters	Values
Learning rate	0.001
No. of network layers	2
No. of neurons per layer	64
Hidden layer activation function	Relu
Output layer activation function	tan
Episode	1500
Optimization algorithm	Adam

The historical reputation value of each HRS is shown in Table G5.

TABLE G5 HISTORICAL REPUTATION VALUES

	HRS1	HRS2	HRS3	HRS4	HRS5
Reputation value	0.9	0.9	0.8	0.7	0.8
	HRS6	HRS7	HRS8	HRS9	HRS10
Reputation value	0.8	0.9	0.9	0.9	0.86

Under Case 5, the preference ranking of seller HRS for buyer HRSs is shown in Table G6, and the preference ranking of buyer HRS for seller HRSs is shown in Table G7. In Table G6, each row corresponds to the preference list of a seller HRS for all buyer HRSs. The numbers in the table represent the specific preference ranking. For seller HRS 1, buyer HRS 10 ranks first in its preference list, followed by HRS 9, HRS 7, and so on. The trading platform executes the many-to-many matching based on this preference ranking. For seller HRS 1, it is most willing to trade its hydrogen with buyer HRS 10.

TABLE G6 SELLER HRS PREFERENCE MATRIX UNDER CASE 5

	HRS5	HRS7	HRS8	HRS9	HRS10
HRS1	2	4	5	3	1

HRS2	2	4	5	3	1
HRS3	2	4	5	3	1
HRS4	2	4	5	3	1
HRS9	2	4	5	3	1

TABLE G7 BUYER HRS PREFERENCE MATRIX UNDER CASE 5

	HRS1	HRS2	HRS3	HRS4	HRS9
HRS5	4	5	2	3	1
HRS6	5	4	2	3	1
HRS7	5	4	2	3	1
HRS8	4	5	2	3	1
HRS10	3	5	2	4	1

Under Case 6, the preference ranking of seller HRS for buyer HRSs is shown in Table G8, and the preference ranking of buyer HRS for seller HRSs is shown in Table G9.

TABLE G8 SELLER HRS PREFERENCE MATRIX UNDER CASE 6

	HRS5	HRS7	HRS8	HRS9	HRS10
HRS1	5	3	4	2	1
HRS2	5	3	4	2	1
HRS3	5	3	4	2	1
HRS4	5	3	4	2	1
HRS9	5	3	4	2	1

TABLE G9 BUYER HRS PREFERENCE MATRIX UNDER CASE 6

	HRS1	HRS2	HRS3	HRS4	HRS9
HRS5	3	1	5	4	2
HRS6	3	1	5	4	2
HRS7	3	1	5	4	2
HRS8	3	1	4	5	2
HRS10	2	1	4	5	3